

Oguz Nurlu

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EDUCATION

Dublin City University

Sept. 2025 – Sept. 2026

MSc in Electronic & Computer Technology (Major in IoT) – First Class Honours (1.1)

Dublin, IE

- **Master's Thesis:** Hardware Acceleration of Multi-Task Convolutional Neural Networks on RISC-V Architectures (TinyKITNet, see Projects)
- **Coursework:** Network Stack Implementation (Linux kernel internals in C), Data Analysis, Machine Learning, Wireless Communications, Security for Edge Networks

TOBB University of Economics and Technology

2021 – 2025

Bachelor of Science in Computer Engineering

Ankara, TR

- **Graduation Project:** Automated Story Generation and Multimodal Rendering Engine
- **Coursework:** Big Data (AWS S3/Lambda, Kubernetes), Parallel Processing, Deep Learning, Machine Learning, Computer Vision

EXPERIENCE

Dublin City University

Jan. 2026 – May 2026

Teaching Assistant & Demonstrator

Dublin, IE

- Served as module demonstrator for EEN1037 (Web Application Development)
- Led weekly lab sessions, mentored students on full-stack architecture, and managed assignment evaluations

STM

Sep. 2024 – Dec. 2024

Full Stack Developer Intern

Ankara, TR

- Contributed to CyThreat, STM's cyber threat intelligence portal, ensuring frontend and backend compatibility
- Enhanced authentication systems (including 2FA), logging mechanisms, and external threat intelligence APIs
- **Technologies:** React, Docker, Django REST Framework, Logstash, Elasticsearch

STM

Jan. 2024 – Apr. 2024

Full Stack Developer Intern

Ankara, TR

- Developed and maintained responsive UI features and backend endpoints for the CyThreat platform
- Optimized database queries and API response latencies for large-scale security incident datasets
- **Technologies:** React, Django, MySQL, Elasticsearch

Jotform

May 2023 – Aug. 2023

Frontend Developer Intern

Ankara, TR

- Developed and optimized user interface components for the Jotform Salesforce integration
- **Technologies:** React, JavaScript, HTML5, CSS3, SOQL

BTK Akademi

Jan. 2023 – Feb. 2023

Instructor

Ankara, TR

- Delivered intensive "Introduction to Programming Using Java" course to 50+ university students
- Designed the curriculum, assignments and sample projects from scratch; published open source on [GitHub](#)

PROJECTS

- Field-Report-Assisted Risk Agent** | *RF-DETR, YOLO, LLM Agents, National Hackathon 2nd Place* 2026
- Won **2nd place nationally** and the ROKETSAN AI Award at ROKETSAN Level Up AI, a national 48-hour hackathon in Türkiye; captained a 6-person team as one of 72 engineers selected from ~2,500 applicants nationwide
 - **Stage 1:** selected the detection models (RF-DETR-M, YOLO26m) and designed and tested the augmentation strategy, including oversampling of rare bus and night images, contributing to a 4-model TTA + WBF ensemble for 4-class aerial vehicle detection at **0.863 val mAP50** (0.809 Kaggle public)
 - **Stage 2:** built the LLM agent pipeline that weighs drone detections, vehicle tracks and potentially misleading field reports; the LLM decides, while a deterministic core does the geo-referencing, track matching and report verification and enforces **8 code-level guardrails** against hallucinated or prompt-injected decisions
 - Added a 15-tool chat agent with on-demand image inspection and streamed answers, and wrote automated tests for the agent, API feed and mobile app (pytest, Flutter unit and widget tests)
 - Built **Karargah Gözü**, a Flutter Android app that replays the day live over SSE, pushes alerts through Firebase Cloud Messaging, and takes Turkish voice questions with spoken answers
- RepForth: Exercise Planner & Tracker** | *Kotlin, Android, Wear OS, Gradle* 2026
- Designed and released a local-first Android training app with a **Wear OS companion**, shipped at v1.0.2 with signed APKs
 - Built as a multi-module Gradle project (core, feature, design-system, build-logic, wear) with baseline profiles for startup performance
 - No account, no backend and no telemetry by design; ships a catalogue of **1,324 exercises** with animations and instructions
 - Fully bilingual in English and Turkish, with neither treated as a translation of the other; published open source with contributor, security and privacy documentation
- Cairn: AI-Verified Habit Tracker** | *Flutter, Dart, Supabase, Riverpod, Drift* 2026
- Built and released a cross-platform habit tracker where every completion is proven by a photo checked by an **AI verifier** before it counts
 - Designed a **local-first** architecture with Drift/SQLite offline storage and Supabase sync, so entries record without a connection and reconcile later
 - Shipped a server-side proof verification service, subscription entitlements, localisation, local notifications, and home screen widgets
- Vision-Language-Action Models for Robotic Manipulation** | *OpenVLA, PyTorch, MuJoCo* 2026 – Present
- Fine-tune and evaluate **OpenVLA** and **OpenVLA-OFT** policies on long-horizon manipulation tasks such as cooking and dishwashing
 - Investigate **modality bias** in vision-language models, measuring how visual attention decays across generated action sequences
 - Extended the stable-worldmodel harness with a GRU-based world model and benchmarked it on MuJoCo cube manipulation
- Varlık Defteri: Property & Investment Manager** | *React, TypeScript, Vite, Tailwind, Django REST* 2026
- Full-stack private asset management system: a React + TypeScript SPA against a Django REST Framework API with SQLite
 - Modelled property types through an abstract base model with inheritance to eliminate duplication across deeds, land, shops and cooperative shares
 - Built local-first and network-isolated by design, with fully protected routes so no UI or API data loads before authentication
- RSVPro: Speed Reading Application** | *React Native, Expo Router, TypeScript, Supabase, Zustand* 2026
- Built an RSVP reader that flashes words at 100–1000+ WPM with Optimal Recognition Point alignment and dynamic pacing on punctuation and word length

- Implemented document import with PDF text extraction, a progress-tracking library, gesture-based scrubbing and speed control, and OLED/sepia themes
- Typed end to end, with Zustand state management, NativeWind styling and Supabase for accounts and sync

TinyKITNet: Hardware-Accelerated Multi-Task CNN | *MSc Thesis, Verilog, RISC-V, gem5, PyTorch* 2026

- Designed and trained a 33.7k parameter INT8 CNN for road perception (classification, bounding box, depth estimation) on KITTI
- Implemented a dedicated Verilog accelerator pipeline and a PicoRV32 soft-core SoC with memory-mapped register interfaces
- Achieved a **4–6× cycle reduction** over out-of-order x86/RISC-V baselines in gem5 under bit-identical integer arithmetic
- Built a single-source-of-truth automated verification harness linking PyTorch QAT, C simulation models, and RTL testbenches

eBPF/XDP DDoS Mitigation | *C, eBPF, XDP, BCC, Python, Linux kernel* 2026

- Wrote an **XDP** program in C, loaded through BCC from Python, dropping fragmented ICMP floods at the driver level before the kernel network stack allocates for them
- Benchmarked it against a **Netfilter** implementation, showing why stateless early drop outperforms stateful fragment reassembly under a Ping of Death style attack
- Deployed and tested on a Raspberry Pi over SSH, satisfying the eBPF verifier's bounds checks on every header access

3D Gaussian Splatting Reconstruction Pipeline | *COLMAP, GLOMAP, gsplat, Splatfacto, viser* 2026

- Built end-to-end scene reconstruction pipelines, running structure-from-motion with **COLMAP** and **GLOMAP** on captured image sets
- Trained and rendered Gaussian Splatting scenes with **gsplat** and **Splatfacto**, applying Gaussian Grouping for object-level segmentation

Browser Extensions & Games | *JavaScript, Unity, Godot, C#* 2023 – 2026

- Published Chrome extensions including a YouTube video summariser and an AI-detector analysis tool
- Developed 2D games in **Unity** and **Godot**, including Nine Bowls (cat puzzle), Re Boot Repair Shop and SlotMatch

SensoryPod: IoT Companion App | *Flutter, Dart, IoT* 2025

- Built a Flutter control app for sensory IoT hardware, covering music and ambient sound playback, smart lighting scenes, and sensor-driven motion control
- Targeted six platforms from one codebase (iOS, Android, Web, Windows, macOS, Linux) with theming and onboarding flows

AI Mood Detector | *React 19, Vite, Tailwind, Deep Learning, Gemini API* 2025

- Full-stack web app performing facial emotion recognition on uploaded images, then generating personalised responses through the Gemini API
- Built the interface in React 19 with Vite, Tailwind and Framer Motion against a Python deep-learning backend

ThalAI: Multimodal Cartoon Episode Engine | *Django, React Native, Docker, Generative AI* 2024 – 2025

- Engineered an end-to-end automated platform generating complete audiovisual episodes for animated series, as my undergraduate graduation project
- Fine-tuned LLM backbones to produce coherent narrative scripts, scene breakdowns and character dialogue
- Architected it as separate services: a Django backend, a containerised LLM training and inference stack, a text-to-speech backend, and a React Native client

ETUPedia | *React Native, Expo, Django REST, PostgreSQL, Docker* 2024

- Designed and published a full-stack social forum and academic directory application for university students
- Scaled to over **300 active users** with real-time feeds, push notifications, and caching; published on the [Play Store](#)

- Implemented full internationalisation with i18next, themed navigation, and animated interfaces; source published on [GitHub](#)

Yanındayım Mobile App | *React Native, Gemini API, Hackathon 2nd Place*

2024

- Won **2nd place** out of 30+ teams in MobileAction's 24-hour hackathon for an elder-care emergency assistant
- Engineered background accelerometer-based fall detection paired with automated emergency contact alerting

SKILLS

Languages : Python, JavaScript, TypeScript, Dart, Kotlin, Java, C, C++, C#, SQL, Verilog, Bash

Web & Mobile : React, React Native, Expo, Flutter, Django REST Framework, FastAPI, REST API design, Server-Sent Events, Firebase Cloud Messaging, Tailwind, NativeWind, styled-components, Zustand, Riverpod, i18next localisation

Databases : PostgreSQL, MySQL, SQLite, Drift, Supabase, Elasticsearch, relational modelling, query and index optimisation

Cloud & Infrastructure : AWS (S3, Lambda), Kubernetes, Docker, Nginx, Linux (Ubuntu and Arch, daily driver), SLURM/HPC, Gradle

AI & Machine Learning : PyTorch, Hugging Face, LLM fine-tuning and prompt orchestration, tool-calling LLM agents with guardrails, Vision-Language-Action models (OpenVLA), Computer Vision, object detection (YOLO, RF-DETR, WBF ensembles), Quantisation-Aware Training, 3D Gaussian Splatting, MuJoCo, MCP-based agent tooling

Systems & Low Level : Linux kernel networking, eBPF/XDP, Netfilter, RISC-V (RV32/RV64, PicoRV32), soft-core SoC bring-up, gem5, Verilog, Vivado

Engineering Practice : Git, pull requests and code review, automated regression harnesses, CI workflows, Agile cross-functional teams, open-source release management

Game Development : Unity, Godot

Spoken Languages : English (Fluent - TOEFL 97/120, IELTS 7.0), Turkish (Native), German (A2)